THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE FULL SYLLABUS PAPER SET 09

Full Syllabus (Chapters 1–13) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Antioxidants are added to food containing fats and oils to prevent: 1

(a) precipitation **(b)** rancidity **(c)** corrosion **(d)** fermentation

2. Bees inject which acid when they sting? 1

(a) methanoic acid **(b)** citric acid **(c)** acetic acid **(d)** oxalic acid

3. Turmeric turns reddish-brown in a: 1

(a) neutral solution **(b)** base

(c) acid **(d)** salt solution of NaCl

4. Stainless steel is an alloy of iron with: 1

(a) chromium and nickel **(b)** tin

(c) lead **(d)** zinc

5. The black coating on silver is: 1

(a) Ag_2S **(b)** $AgCl$ **(c)** Ag_2O **(d)** $AgNO_3$

6. For the soap molecule, Student A says its ionic end is hydrophilic; Student B says the hydrocarbon end is hydrophilic. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

7. Diamond is hard because: 1

(a) it has layers

(b) it is ionic

(c) each C is bonded to four others in a 3-D network

(d) it has free electrons

8. Which enzyme converts fats into fatty acids and glycerol? 1

(a) lipase **(b)** amylase **(c)** pepsin **(d)** trypsin

9. Hypothalamus releasing factors stimulate the: 1

(a) pituitary **(b)** testes **(c)** pancreas **(d)** kidney

10. A student states: “Oral contraceptive pills change the hormonal balance to prevent eggs being released.” The correct response is: 1

(a) Correct **(b)** Incorrect

(c) Only with copper-T **(d)** Cannot be determined

11. For the testes, Student A says they produce testosterone; Student B says they produce sperms. Then: 1

(a) Both are correct **(b)** Only Student A is correct
(c) Only Student B is correct **(d)** Both are wrong

12. Number of different gametes produced by RrYy is: 1

(a) 1 **(b)** 4 **(c)** 8 **(d)** 2

13. The Y chromosome is found in: 1

(a) males only **(b)** females only **(c)** both **(d)** neither

14. Magnification of a lens is defined as: 1

(a) f/u **(b)** u/v **(c)** $h'/h = v/u$ **(d)** $-v/u$

15. Cataract is a condition in which the: 1

(a) cornea becomes curved **(b)** retina detaches
(c) eye lens becomes milky and cloudy **(d)** pupil is blocked

16. If the potential difference across a resistor is doubled, the current becomes: 1

(a) half **(b)** double **(c)** four times **(d)** same

17. Equivalent resistance of 10 Ω and 15 Ω in parallel is: 1

(a) 25 Ω **(b)** 12.5 Ω **(c)** 6 Ω **(d)** 5 Ω

18. The force on a conductor is reversed when: 1

(a) the length is doubled **(b)** the wire is thicker
(c) the direction of current is reversed **(d)** current is increased

19. The amount of solar energy captured by plants is approximately: 1

(a) 100% **(b)** 1% of the sunlight falling on leaves
(c) 10% **(d)** 50%

20. In the food chain grass → deer → lion, the lion is a: 1

(a) herbivore **(b)** decomposer
(c) tertiary/secondary consumer (carnivore) **(d)** producer

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. What is corrosion? Give one example. 2

22. While diluting an acid, why is it recommended that the acid be added to water and not water to the acid? 2

23. Name the two components of the transport system in plants. 2

24. Why is the use of iodised salt advisable? 2

25. Why is DNA copying essential for reproduction? 2

26. What are the two copies of a gene in a pair called? What happens to them during gamete formation? 2

27. State the two laws of reflection of light. 2

28. Why does a compass needle get deflected near a bar magnet? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Classify with reason: 3

(i) $2\,KI + Cl_2 \longrightarrow 2\,KCl + I_2$

(ii) $2\,H_2O_2 \longrightarrow 2\,H_2O + O_2$

(iii) $C + O_2 \longrightarrow CO_2$

30. Why do ionic compounds have high melting and boiling points? Give two examples with their melting points. 3

31. Why are hormones called chemical messengers? Give two examples of their role. 3

32. Why is sexual reproduction more beneficial than asexual? Give three reasons. 3

33. How is the eye able to see both near and distant objects? 3

34. An electron enters a magnetic field at right angles. Explain its path and the rule to find the force. 3

35. What is the pH scale? How is pH related to the concentration of H^+ ions? Why does a farmer add quicklime to his field? 3

OR

Equal lengths of magnesium ribbon are put in test tubes A (HCl) and B (acetic acid) of the same concentration. In which tube is fizzing more vigorous and why? 3

36. What is the 10% law? Explain with an example of a food chain. 3

OR

Explain how energy is transferred in a food chain of grass → rabbit → fox, using the 10% law (take 10 000 J at grass). 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) An object 5 cm high is placed 25 cm in front of a concave mirror of focal length 10 cm (describe the diagram). 5

(ii) Find the image distance and height.

(iii) State its nature.

OR

Case study: A doctor prescribes a lens of power +2.0 D. 5

(i) Is the lens converging or diverging?

(ii) Find its focal length.

(iii) If an object is placed 25 cm from this lens, find the image distance.

(iv) Nature of the image?

38. (i) What are covalent compounds? 5

(ii) Why are they poor conductors of electricity and have low melting points?

(iii) Draw the electron-dot structure of H_2O and NH_3.

OR

(i) Differentiate between soaps and detergents (any three points). 5

(ii) Why do detergents cause water pollution?

(iii) What is hard water?

39. (i) Explain Joule's law of heating. 5

(ii) State two applications of heating effect.

(iii) An electric heater draws 5 A from a 220 V line. Find its power and energy used in 2 h.

OR

(i) What are the disadvantages of series connection for domestic circuits? 5

(ii) Three bulbs of 4 Ω, 8 Ω and 8 Ω are connected in parallel across 8 V. Find the total current and current in each.

40. (i) Explain the different ways glucose is broken down in organisms with a flow chart. 5

(ii) Where do these processes occur?

OR

Case study: Lymph is a fluid that leaks from capillaries into intercellular spaces. 5

(i) How does lymph differ from blood?

(ii) Name two functions of lymph.

(iii) What are the components of blood?

(iv) What is the function of platelets?

END OF QUESTION PAPER **Total: 80 Marks**

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